

Howard Wen

(403) 890-5086 | hwen.hba2027@ivey.ca | in: [howard-wen04](#)

EDUCATION

Western University & Ivey Business School

London, ON

BESc in Mechatronic Systems Engineering & Honours Business Administration

Expected Apr 2027

- **Engineering:** 3.96 / 4.0 (91.5%); Dean's Honor List, awarded \$12K+ in merit-based scholarships
- **Business:** 3.80/4.0 (85.7%); Dean's Honor List, awarded \$14K+ in merit-based scholarships

WORK EXPERIENCE

Arthur B. McDonald Institute – Queen's University

Kingston, ON

Undergraduate Researcher – Dark Matter Detection

May 2026 – Aug 2026

- Built an integrated slow-control and DAQ system for a liquid-argon cryostat, unifying 8+ instrumentation subsystems (cryogenic RTDs, mass-flow control, LN₂ delivery, and vacuum) into one interface with real-time monitoring, remote operation, and an automated analysis pipeline, enabling stable unattended detector R&D
- Re-implemented Windows-only vacuum and residual-gas-analyzer control as a Linux/MIDAS interface, adding interlock logic across the turbopump and RGA filament to prevent equipment damage during remote operation

Boston Consulting Group (BCG)

Calgary, AB

Summer Associate

Jun 2025 – Aug 2025

- Conducted analysis for a strategy initiative with a major North American midstream operator on commercial levers (e.g., digital & AI, trading, LCPM, cost) and portfolio strategy to address a \$XB near-term EBITDA gap
- Modelled expected capital expenditure on next decade North American midstream infrastructure by basin, commodity and value chain segment; analysis on data, headwinds, and opportunities in BCG-published article

Northern Hail Project

Olds, AB

Field Operations & Instrumentation Intern

May 2024 – Aug 2024

- Conducted severe hailstorm interceptions to collect hailstone data via UAVs and GIS software, while maintaining and calibrating field instrumentation to ensure operational reliability in extreme weather conditions
- Led prototype development on a hail velocity measurement instrument by designing weather resistant enclosures and programming sensor software, providing new data collection techniques with 30k+ data points

EXTRACURRICULAR EXPERIENCE

Arcadian Research & Contracting

London, ON

President

Sep 2024 – Present

- Managed ARC, a 5-person engineering organization delivering 3 completed projects to academics and researchers ranging from software to instrumentation at Western University, expediting key research goals

Select Project Experience (\$10k+ total value):

- **Rugged Radome Hail Instrument for Canadian Severe Storms Laboratory (CSSL)**
 - Designed and built four rugged UHMW polyethylene radome enclosures using SolidWorks, conducting FEA simulations and drop tests; delivered to a multimillion-dollar US severe hailstorm research initiative
 - Developed IoT software for a 4G-enabled Raspberry Pi controller running C++ and Python-based embedded systems software collecting and uploading processed signal data to a remote frontend dashboard
- **USA Farm Perimeter Software for Free Appropriate Sustainability Technology Group (FAST)**
 - Created modular software that analyzes GIS data from NASS's crop sequence boundaries, decomposing it into separate shapefiles by state, and calculating total raw and dissolved perimeter of polygons for FAST

Entrepreneurship Through Acquisition (ETA) Club

London, ON

Research Analyst - Indirect Infrastructure Research

Nov 2025 – Present

- Mapped upstream/downstream value chains across three prioritized Canadian nation-building megaprojects to identify and screen 130+ candidate firms positioned as third-order beneficiaries of federal infrastructure capex

SKILLS, ACTIVITIES & INTERESTS

Skills: Python, React.js, C++, MIDAS, Git, Linux, C, KiCAD, MATLAB, MongoDB, SolidWorks, 3D Printing

Activities: IEEE VP Ops, Ivey Consulting Club Mentor, Engineering O-Week Leader, Western AI VP Finance

Interests: Skiing, Chopin, Experimental Cooking